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Optimum utilization of grid-connected renewable energy sources using multi-year module in Tunisia

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Abstract This study aims to raise awareness of renewable energies' importance from an economic and environmental perspective and provide reference data for the investment decision in Tunisia's renewable energy. It focuses on the techno-economic analysis of a grid-connected photovoltaic-wind power system to supply residential load in 26 cities in Tunisia using the multi-year module. Three types of small wind turbines are studied to select suitable wind turbines for each site based on the capacity factor concept. HOMER simulation software is used to design and determine the feasibility of the systems. The optimal configuration is obtained at each site of the country in terms of the lowest net present cost and CO_2 emissions using the Wes-tulipo wind turbine. The obtained results show that wind energy is suitable for most cities. It is even more profitable and competitive than photovoltaic (PV) in northern and central Tunisia. In contrast, the PV system has great potential for development in the southern regions of the country. The economic benefit of the renewable energy project for each site is between 11 and 16 years during the 25-years project lifetime, proving its economic attractiveness. Furthermore, this study's sensitivity analysis proves that the PV system becomes competitive if its capital cost is reduced to less than 57% of its current capital cost.

1. Introduction

One of the biggest challenges of the 21st century is meeting growing global energy needs while tackling climate change and greenhouse gas emissions. Currently, renewable energies represent the best solution as a clean and inexhaustible resource. Therefore, the development of renewable energies and the promotion of their use are essential to ensure our energy independence and meet the climate challenge. The share of renewable energy in global electricity production reached 24.9% in 2018. It consists of 63% hydropower, 18% wind energy, 9% solar energy, 8% bioenergy, and 1% geothermal energy [1]. According to the latest statistical report from the International Renewable Energy Agency, renewable energy accounted for 75% of new power generation facilities in 2019 [2]. Despite this, the share of renewable energy in global electricity production remains low. However, some countries reached very satisfactory levels. For example, Denmark and Uruguay generated 60% and 33% of their electricity needs from renewable energies in 2019 [3].

Government policies play an essential role in the development and use of renewable energy technologies. As of 2017,

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84% of countries had a legal and regulatory framework to support the deployment of renewable energies [4]. Tunisia is one of the developing countries that has given great importance to renewable energy as an essential part of its energy conservation policy. Tunisia has adopted an energy transition policy that aims to increase the share of renewable energies in electricity production from 3% currently to 30% in 2030. To achieve this objective, Tunisia established a regulatory framework by issuing the 12/2015 law related to electricity production from renewable energies. It consists of three regulatory systems: self-consumption, authorizations via calls for projects, and concessions by calls for tenders [5]. Per the provisions of Article 11 of the law mentioned above, any Tunisian Company of Electricity and Gas (STEG) customer can produce electricity from renewable energy sources for his consumption and benefit from the right to sell the surplus of electricity produced exclusively to the STEG [6].

The electrical grid plays an essential role in the reliability and economic feasibility of hybrid renewable energy systems. Since storage systems represent a significant portion of the total system cost, they can be omitted from the grid-connected renewable energy systems [7]. In this case, the energy produced from renewable resources feeds the load, and the excess energy is sent to the electrical grid. When renewable energy resources are insufficient, the electrical grid should cover the load. Many studies have investigated the techno-economic feasibility analysis of grid-connected renewable energy sources. Makbul et al. [7] used the net present cost, unmet load, the fraction of renewable electricity, excess electricity, and carbon dioxide emissions for the optimal sizing of a grid-connected PV system in Makkah, Saudi Arabia, using HOMER software. They concluded that the total NPC of the system decrease when the inverter size is downsized to 68% of the PV nominal power. In Ref. [8], Moslem et al. performed a techno-economic analysis of a grid-connected PV-wind power system for a residential urban area. Their results show that the optimal system consists of four wind turbines and a power grid with an initial investment of 440 thousand dollars. The PV power system can be involved when the average wind speed is lower than 3 m/s. Sultan et al. [9] proposed an improved artificial ecosystem optimization technique for the optimal design of a grid-dependent and off-grid hybrid renewable energy system through minimizing the cost of energy and the loss of power supply probability. Shirzadi et al. [10] studied the optimal sizing of an integrated renewable energy system for isolated and grid-connected microgrids and compared their economic and technological performance. The results show that using the grid as a backup with renewable energy sources could reduce electricity costs by about half compared with Quebec's grid electricity price. In Ref. [11], Weiwu et al. performed a techno-economic analysis of grid-connected renewable power systems, including wind turbine systems, PV power systems, and biogas power systems in five climatic regions of China. Based on technical, environmental, economic, and comprehensive evaluation indexes, the analysis conducted to provide a national energy plan for the investment decision in renewable energy in China. Zhang et al. [12] simulated a grid-connected PV battery system under the dynamic electricity tariffs to carry out a techno-economic analysis of different system combinations. They considered the external factors including seasonal load variance, battery self-discharge, and weather conditions. Uddin and Islam [13] presented an optimization method and energy management strategy of a hybrid grid-connected wind-PV power system for multi house. HOMER software is used to optimize the system configuration and a fuzzy logic controller is applied to control the management strategy. This study found an annual cost savings of 11.3% and a reduction in CO_2 emissions by 99%. Barakat et al. [14] used a multi-objective optimization for the optimal design of a grid-connected PV-wind hybrid system to feed a load in the Egyptian countryside. This study has considered three objectives, including the minimization of the cost of energy and loss of power supply probability while maximizing the renewable energy fraction of the system. In Ref. [15], Diab et al. presented a modified farmland fertility optimization algorithm for optimal sizing of a grid-connected PV-fuel cell-wind power system by minimizing energy cost while satisfying the operational constraints. Ndwali et al. [16] proposed a multi-objective sizing method of a grid-connected PV system without energy storage to minimize the energy purchased from the electrical grid and then total life cycle cost while maximizing the reliability of the hybrid system.

In a techno-economic analysis of grid-connected renewable energy sources, some parameter changes occur over the project lifetime, which affects the reliability of the optimization results. However, published studies did not consider these parameters, such as diesel prices, grid prices, PV panel degradation, battery degradation, and load growth. In addition, greenhouse gas emissions are neglected or not taken into account. In this context, this work is situated. It focuses on the techno-economic analysis of grid-connected PV-wind power systems using multi-year module inputs in 26 cities in Tunisia. Based on the economic and environmental evaluation, this study is conducted to provide a national energy plan and reference data for the investment decision in Tunisia's renewable energy. Moreover, a sensitivity analysis has been included to reveal the effect of cost change on optimization results.

The contributions of the paper are summarized as follows:

- The economic and environmental factors in the optimization strategy of the grid-connected renewable energy sources are considered.
- Selecting a suitable wind turbine for a given site ensures both technical feasibility and financial competitiveness in terms of high production and low electricity costs.
- Using the grid as a backup with renewable energy sources reduces the total net present cost of the system and the payback period.
- Provide some valuable reference data for the investment decision in renewable energy in Tunisia.

This paper is organized as follows: Section 2 presents the hybrid power system with all its components, the load profile, the meteorological data, and the multi-year module inputs. Section 2 also describes the methodology adopted for the optimal design of grid-connected renewable energy sources. Section 3 discusses the optimization results and sensitivity analysis. Finally, the conclusion is presented in Section 4.

2. Optimal design of grid-connected renewable energy sources

Due to the continuous increase in Tunisia's electricity prices, consumers must rethink how to produce and use cheap and sustainable electricity. Renewable energy is the key to meet this challenge, especially as Tunisia has excellent renewable resources such as solar and wind energy. This section presents a techno-economic analysis of a grid-connected hybrid renewable energy system for a typical house in 26 cities in Tunisia.

2.1. Hybrid power system

Fig. 1 shows the block diagram of the proposed HPS. The PV array supplies the direct current (DC) bus while the wind generator is joined to the alternating current (AC) bus through a converter. The DC power generated by the PV array is converted into AC by the inverter. The load is considered a household where it needs an alternating current. The energy produced from the PV panels and wind generators feeds the load. The electrical grid is used instead of storage systems since it serves almost the entire population in Tunisia. The excess energy from renewable sources is sent to the electrical grid through the bi-directional meter. When renewable energy resources are insufficient, the electrical grid should cover the load.

Pre-feasibility analysis and sizing of the integrated HPS play a critical task in deciding the system's reliability and economy. The pre-feasibility study is based on the weather data and the load required for the specific site.

2.2. Meteorological data and load profile

2.2.1. Load profile

In this study, a medium-size house of a Tunisian family consisting of six persons is chosen for evaluation. The daily electric load of the house is collected from the Tunisian Company of Electricity and Gas. The monthly load profile is illustrated in Fig. 2. It shows that the peak load is about 2.63 KW with an energy consumption of 18.94 KWh/day observed in June.

2.2.2. Meteorological data

NASA's Surface Meteorology and Solar Energy database is referred to obtain the temperature, the solar radiation, and the wind speed data at 50 m above the earth's surface for all country sites.

Tunisia has the advantage of being exposed to a total insulation period of 3500 h/year and 350 sunny days per year. The solar radiation intensity varies between 4.5 kWh/m^{-2} d^{-1} in the north and 7 kWh/m^{-2} d^{-1} in the south [17], see Fig. 3.

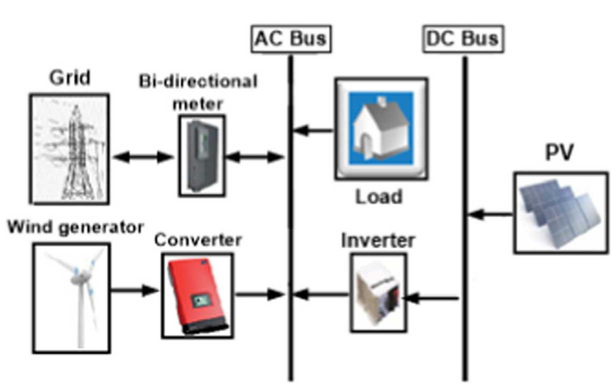

Fig. 1 Proposed hybrid power system.

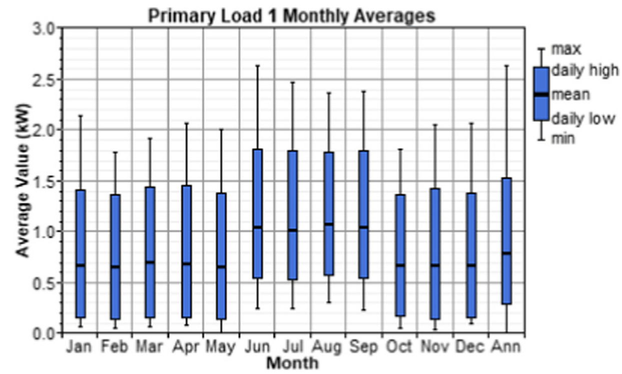

Fig. 2 Monthly average load profile.

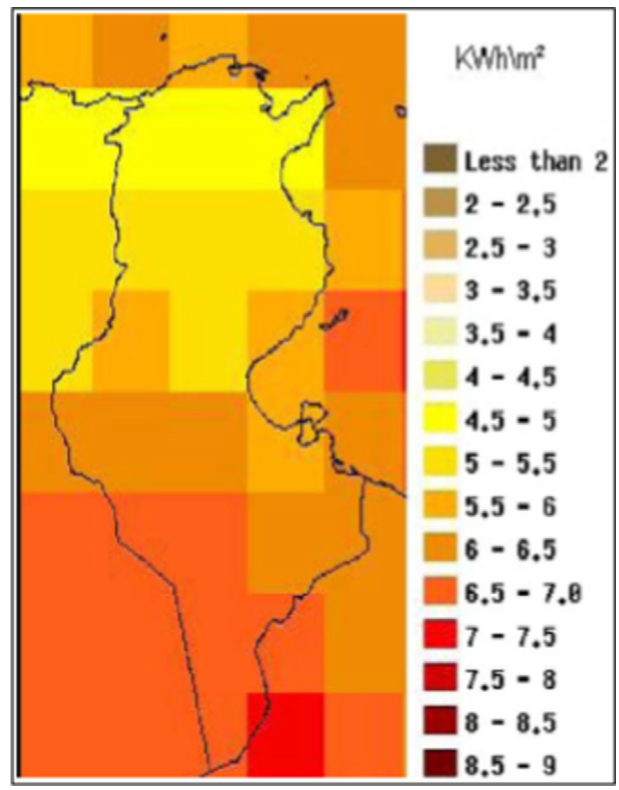

Fig. 3 Distribution of the solar radiation.

Tunisia also has a significant wind energy potential estimated at 8GW [18]. The regions with high wind potential are mainly found in the northern part of Tunisia and some areas in the center and south. Fig. 4 illustrates the map of the annual spatial distribution of mean wind speed over the country [19].

2.3. Hybrid system components

2.3.1. PV panel

The output power of the PV panel can be calculated using the following equation [20]:

$$P_{pv} = Y_{pv} \times f_{pv}\left(\frac{G_T}{G_{T,STC}}\right)[1 + \alpha_p \times (T_c - T_{c,STC})] \tag{1}$$

Fig. 4 Annual mean wind speed.

where f_{pv} is the PV derating factor (%), Y_{pv} is the rated capacity of the PV array (kW), G_T is the solar radiation incident on the PV array in the current time step (kW/m^2),$G_{T,STC}$ is the incident radiation at standard test conditions (1 kW/m^2), $T_{c,STC}$ is the PV cell temperature under standard test conditions (25 °C), α_p is the temperature coefficient of power (% /°C) and T_c is the PV cell temperature (°C), it's given by the following equation [20]:

$$T_c = \frac{T_a + (T_{c,NOCT} - T_{a,NOCT}) \times \left(\frac{G_T}{G_{T,NOCT}}\right)\left[1 - \frac{\eta_{mp,STC} \times (1 - \alpha_p \times T_{c,STC})}{\tau \times \alpha}\right]}{1 + (T_{c,NOCT} - T_{a,NOCT}) \times \left(\frac{G_T}{G_{T,NOCT}}\right)\left(\frac{\alpha_p \times \eta_{mp,STC}}{\tau \times \alpha}\right)} \quad (2)$$

where $T_{c,NOCT}$ is the nominal operating cell temperature (°C), T_a is the ambient temperature (°C), $G_{T,NOCT}$ is the solar radiation at which the NOCT is defined (0.8 kW/m^2), $\alpha$is the solar absorptance of the PV array (%), $T_{a,NOCT}$ is the ambient temperature at which the NOCT is defined (20 °C), τ is the solar transmittance of any cover over the PV array (%), and $\eta_{mp,STC}$ is the maximum power point efficiency under standard test conditions, it is given by [20]:

$$\eta_{mp,STC} = \frac{Y_{pv}}{A_{pv} \times G_{T,STC}} \quad (3)$$

where A_{pv} is the surface area of the PV panel (m^2).

The derating factor is used to calculate the drop in the PV output power in real operating conditions. This reduction can be due to wiring losses, shading, snow cover, aging, and so on. It is assumed to equal to 80% in this study. The technical specifications of the chosen PV panel are summarized in Table 1.

Table 1 **Specification details of PV module [21].**

Parameters	Specifications
Model	TITAN-12–50
Rated power	50 W
Maximum power voltage	17.2 V
Maximum power current	2.9 A
Open circuit voltage	21 V
Short circuit current	3.2 A
Temperature coefficient (Voc)	−0.38%/°C
Temperature coefficient (Isc)	+0.1%/°C
Temperature coefficient of power	−0.47%/°C
Maximum system voltage	600 V
Capital cost	100$
Replacement cost	100$
Maintenance cost	0$
Derating factor	80%
Ground Reflectance	20%
Lifetime	25 years

2.3.2. Wind turbine

The output power of a wind turbine in each time step can be calculated using a three-step process [20]:

First, the wind speed at the hub height of the wind turbine is adjusted using the logarithmic profile:

$$\frac{v(z_{hub})}{v(z_{anem})} = \frac{\ln(z_{hub}/z_0)}{\ln(z_{anem}/z_0)} \quad (4)$$

where z_{anem} is the anemometer height (m), z_{hub} is the hub height of the wind turbine (m), $v(z_{anem})$ is the wind speed at anemometer height (m/s), $v(z_{hub})$ is the wind speed at the hub height of the wind turbine (m/s), and z_0 is the surface roughness length (m).

Second, reference is made to the wind turbine power curve, under standard temperature and pressure conditions, to calculate the power output for corresponding wind speed, which is found in the first step.

Finally, the output power predicted by the power curve from the second step should be multiplied by the air density ratio given by the following equation:

$$\frac{\rho}{\rho_0} = (1 - \frac{B \times Z}{T_0})^{\frac{g}{RB}}(\frac{T_0}{T_0 - B \times Z}) \quad (5)$$

where B is the lapse rate (0.00650 K/m), ρ is the air density (kg/m3), T_0 is the standard temperature (288.16 K), Z is the altitude (m), R is the gas constant (287 J/kgK), and g is the gravitational acceleration (9.81 m/s2).

The choice of a wind turbine is an essential operation. It depends on the aerogenerator specifications and the implantation site. Therefore, a comprehensive match of wind turbine generators with sites is required to ensure both technical feasibility and financial competitiveness in terms of high production and low electricity costs. One way to match wind generators to sites is through the determination of the capacity factor. Wind turbines with higher capacity factors are favored. The capacity factor is the average power output of the wind turbine over a period (usually a year) divided by its rated peak power. The average power output is the total electricity generated divided by the number of hours of that period [24].

This study considered three types of small wind turbines to choose the suitable wind turbine for each site based on the capacity factor concept. The first type is the 2.5 kW rated power Wes tulip wind turbine. It is a three-blade Dutch wind turbine made by Wind Energy Solutions Company. Fig. 5 and Table 2 show its power curve and its specification details, respectively. The second type of wind turbine is the Fortis passaat with a 1.4 kW nominal power output. Fig. 6 and Table 3 show the power curve and specification details of this wind turbine. The third type is the 2 kW rated power Hummer H3.1 wind turbine. Its power curve and its specification details are available in Fig. 7 and Table 4, respectively. The first wind turbine produces alternating current and directly connecting to the AC bus, but the two other wind turbines need a converter because they generate DC power.

2.3.3. The electrical grid

The load is connected through a bi-directional meter to the electrical network (220v, 50 Hz). The latter receives extra energy from renewable sources and covers the shortage of energy. At the end of the billing period (monthly or annually), the net amount purchased (electricity purchases minus sales) is calculated. If the value is negative, the utility pays you according to the sellback price.

Tariffs of electricity produced from renewable sources and sold to STEG is shown in Table 5 [28]. The cost of buying electricity from the grid is illustrated in Table 6 [29].

2.3.4. Inverter and bi-directional meter

The inverter is a power electronic device used to convert PV output power from DC to AC. The average price of the inverter is estimated at 400$/kW, and its lifetime is up to 15 years, with 95% efficiency.

The bi-directional meter is used to record the total electricity received from the grid and fed into the grid. The metering system is supplied and installed by STEG, which ensures its periodic maintenance.

2.4. Multi-year inputs

The multi-year module allows us to model the parameter changes that occur over the project lifetime as diesel prices, grid prices, PV panel degradation, battery degradation, and load growth. In other words, it allows you to input year-by-

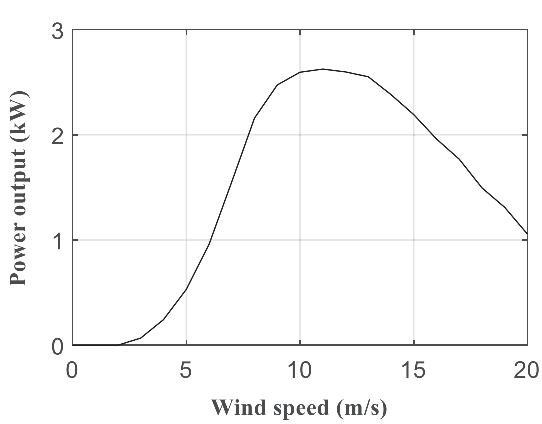

Fig. 5 Power curve of Wes tulipo.

Table 2 Specification details of the first wind turbine [22,23].

Parameters	Specifications
Manufacturer	Wind Energy Solutions
Model	Wes tulipo
Rated power	2.5 KW
Rated speed	8.5 m/sec
Cut-in speed	3 m/sec
Cut-out speed	20 m/sec
Tower height	30 m
Rotor diameter	5 m
Swept area	19.6 m^2
Maximum System Voltage	400VAC
Generator type	asynchronous generator
Capital cost	5000$
Replacement cost	5000$
Maintenance cost	75$
Lifetime	20 years

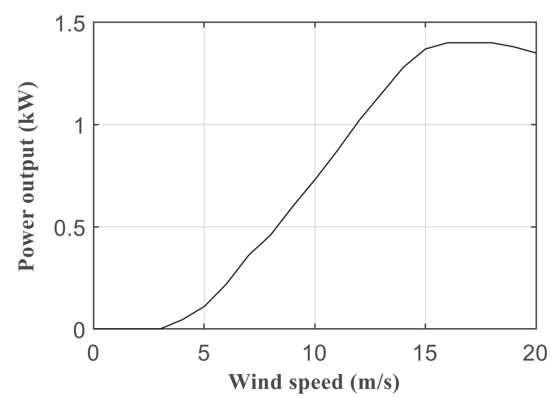

Fig. 6 Power curve of Fortis passaat.

Table 3 Specification details of the second wind turbine [22,25].

Parameters	Specifications
Manufacturer	Fortis Wind Energy
Model	Fortis passaat
Rated power	1.4 KW
Rated speed	16 m/sec
Cut-in speed	2.5 m/sec
Cut-out speed	25 m/sec
Tower height	30 m
Rotor diameter	3.12 m
Swept area	7.64 m^2
Maximum System Voltage	350VDC
Generator type	Synchronous, PMG
Capital cost	3900$
Replacement cost	3900$
Maintenance cost	58.5$
Lifetime	20 years

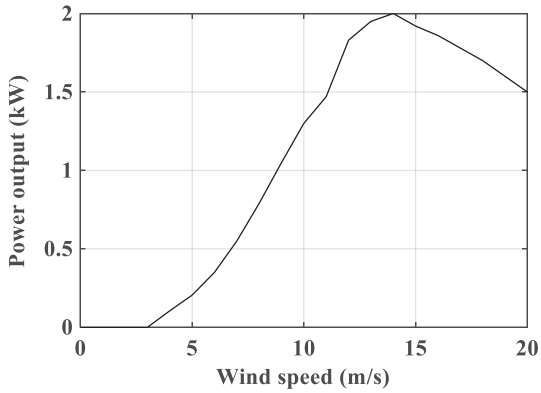

Fig. 7 Power curve of Hummer H3.1.

Table 4 Specification details of the third wind turbine [26,27].

Parameters	Specifications
Manufacturer	Hummer
Model	h3.1
Rated power	2 KW
Rated speed	14 m/sec
Cut-in speed	3 m/sec
Cut-out speed	25 m/sec
Tower height	30 m
Rotor diameter	3.1 m
Swept area	8.04 m^2
Maximum System Voltage	55VDC
Generator type	Synchronous, PMG
Capital cost	4150$
Replacement cost	4150$
Maintenance cost	62.25$
Lifetime	20 years

Table 5 Price of electricity sold to the grid.

Times	Price in $/kWh
day	0.04
peak morning summer	0.063
peak evening	0.058
night	0.03

Table 6 Power price from the electrical grid.

Consumption in kWh	Price in $/kWh
[1 200]	0.06
[201300]	0.075
[301 500]	0.11
> 500	0.142

year anticipated percentage changes in some parameters over the lifetime of the project [30]. In this study, the simulation is performed by considering the following phenomena:

– The PV degradation: PV modules are designed to operate reliably for 25 years. But the PV panel is exposed to external and internal stresses that can drive its degradation, resulting in the reduction of its durability. Some studies have proved that the PV module's median degradation rate is between 0.5% and 1%/year [31,32]. For that, the degradation rate is considered at 0.5%/year over the 25-years project lifetime.
– The grid price: In Tunisia, the electricity sector is subsidized by the government because the selling prices do not cover the costs of electricity production and distribution. In order to reduce subsidies, the government has planned an increase in electricity prices every year. By 2014, the cost of electricity has increased by 10%. After that, the annual increase in electricity prices became fluctuating between 3 and 10%, according to international gas prices and the country's financial balance [33,34,35]. For that, the grid price increment is assumed to be 5%/year over the 25-years project lifetime.

For every year, grid prices considered in the simulation are that given in Tables 5 and 6 multiplied by the multiplier given in Table 7.a. PV power considered in simulation for every year is that produced by the PV array, under insolation and temperature condition for a given site, multiplied by the multiplier given in Table 7.b.

2.5. Optimization methodology

HOMER software developed by National Renewable Energies Laboratory is used in this study to find the optimal configuration of a hybrid power system (HPS) for the chosen site. The software performs the grid-connected renewable power system analysis by simulating the system operation and cost evaluation during the project's lifetime. HOMER's optimization algorithms generate optimizations results for dozens of feasible configurations and assort them from the lowest to the highest total net present cost (TNPC).

The project lifetime is estimated at 25 years, and the annual interest rate is fixed at 8%. The following constraints are considered in this study:

– Maximum annual capacity shortage equal to 0%.
– Minimum renewable fraction equal to 60%.

HOMER requires three different inputs to define the excess operating capacity. The three parameters are as below:

– Operating reserve as a percentage of load equal to 10%.
– Operating reserve as a percentage of solar power output equal to 5%.
– Operating reserve as a percentage of wind power output equal to 10%.

In order to reduce the computation time, the maximum number of wind turbines and the maximum number of PV

Table 7 Multi-year variable detail.

a. Grid price

Year	1	2	3	4	5	6	7	8	9	10	11	12	13
Multiplier	1	1.05	1.102	1.157	1.215	1.276	1.34	1.407	1.477	1.551	1.628	1.710	1.795
Year	14	15	16	17	18	19	20	21	22	23	24	25	
Multiplier	1.885	1.979	2.078	2.182	2.292	2.406	2.526	2.653	2.785	2.925	3.071	3.225	

b. PV degradation

Year	1	2	3	4	5	6	7	8	9	10	11	12	13
Multiplier	1	0.995	0.99	0.985	0.98	0.975	0.97	0.965	0.96	0.955	0.95	0.945	0.941
Year	14	15	16	17	18	19	20	21	22	23	24	25	
Multiplier	0.936	0.931	0.926	0.922	0.917	0.912	0.908	0.903	0.899	0.894	0.89	0.885	

panels to be considered in the search space of HOMER are calculated by the following two expressions:

$$Nw = \frac{P_{load}}{Pw} \tag{6}$$

$$Npv = \frac{P_{load}}{Ppv} \tag{7}$$

where Ppv and Pw are the average hourly power produced by the photovoltaic module and the wind generator during the first year, respectively. P_{load} is the average hourly power demand by the load during the year.

For every system configuration, HOMER performs an hourly time series simulation every year in order to determine the operational characteristics such as annual energy production, annual energy purchased and sold to the grid, CO_2 emissions, TNPC, cost of energy, annual load served, excess electricity, renewable fraction, and so on. Fig. 8 depicts the schematic of the feasibility process for each configuration over one year. The required input data includes technical and economic data of all components, electrical loads, meteorological data, and technical constraints. The electrical grid covers the load when renewable energy resources are insufficient and receives the excess energy generated. Iterative simulation of each configuration is repeated every year over the 25-years project lifetime by considering multi-year inputs to decide its feasibility if constraints are respected.

The next step after classifying all feasible configurations based on TNPC is to find the optimal system configuration. The optimal design objective is to minimize the TNPC of the system while reducing CO_2 emissions. Since the complementary nature of solar and wind energy resources creates a stable energy supply, it reduces the need for the electrical grid and then decreases CO_2 emissions. For that, it is better to choose an optimal configuration consisting of the two renewable energy sources. This configuration should be economically feasible, has an attractive payback period, and less CO_2 emissions. The strategy for selecting the optimum configuration is as follows:

- If the configuration with the lowest TNPC (TNPC_L) is hybrid (wind + PV) and presents the lowest CO_2 emissions, it will be considered the optimal configuration. Else, suppose this configuration is hybrid but does not have the lowest CO_2 emissions. In that case, another hybrid configuration with less CO_2 emissions and a TNPC lower than 110% of the TNPC_L should be considered. If it does not exist, the one with the lowest TNPC is kept as the optimal configuration.
- If the configuration with the lowest TNPC consists only of a PV system or a number of wind turbines greater than 1, a hybrid configuration (wind + PV) with a TNPC lower than 110% of TNPC_L and the lowest possible CO_2 emissions should be considered. If it does not exist, the one with the lowest TNPC is kept as the optimal configuration.
- Otherwise, the optimal configuration is the one with the lowest TNPC.

The cost limit was chosen equal to 110% of TNPC_L because the main objective of the optimization is economical.

3. Results and discussions

3.1. Optimization results

This study's main objective is the size optimization of the grid-connected hybrid renewable energy system for supplying a residential load in the 26 sites of the country. The capacity factor of the three wind turbines over a year for all sites is given in Fig. 9. We notice that the WES Tulipo wind turbine's capacity factor is higher than that of the two other turbines. Therefore, the optimum choice of the appropriate wind turbine for all sites is the WES Tulipo turbine.

Based on the wind speed in the Thala region, a comparison of the power generated from the three wind generators is made, Table 8. The average power output and the total energy output over the year for the WES Tulipo wind turbine are higher than that of the two other wind turbines. This result proves the importance of choosing the right wind turbine for a given site.

Table 9 shows the optimal configuration of the HPS for each site in terms of the lowest net present cost and CO_2 emissions using the WES Tulipo wind turbine. The optimization results show that wind energy is suitable for most cities and even more profitable and competitive than PV in northern and central Tunisia. In contrast, the PV system has excellent potential for development in the southern regions, except the

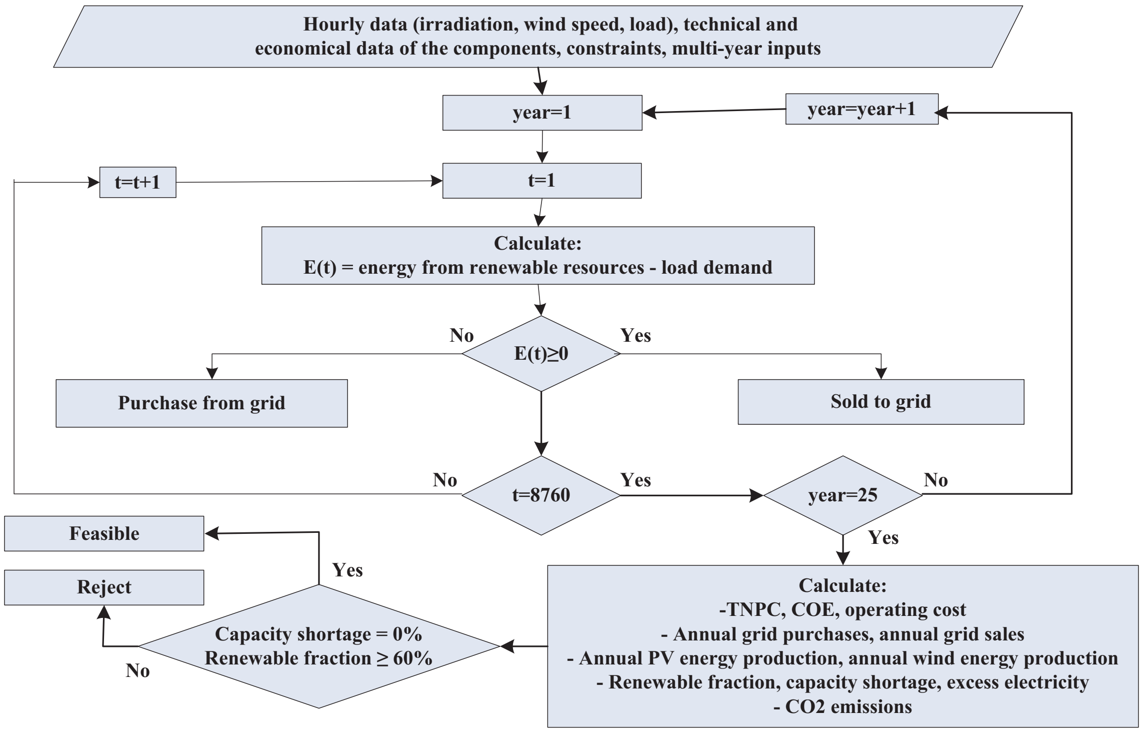

Fig. 8 Flowchart summarizing the feasibility process for each configuration.

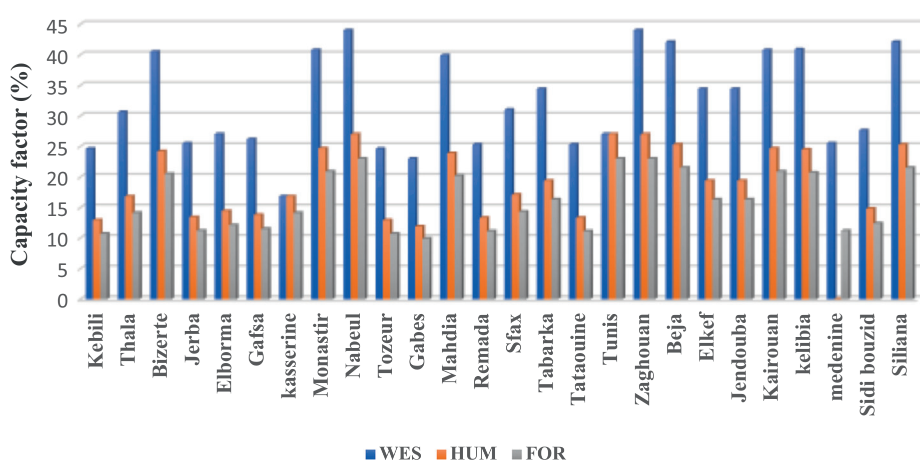

Fig. 9 Capacity factor of the three wind turbines for the 26 sites of the country.

Table 8 Output power of the three wind turbines over the year.

Model	WES Tulipo	Hummer H3.1	FORTIS PASSAAT
Mean output (kW)	0.804	0.336	0.197
Total production (kWh)	7039	2946	1728

Sfax region, where wind energy is more economical than PV energy.

These results can be confirmed by comparing the levelized cost of energy (LCE) from the wind turbine with that from the PV array for the country's 26 cities. Indeed, the LCE of the wind turbine for the northern and central cities ranges from 0.061 \$/kWh to 0.073 \$/kWh. For the same cities, The LCE of the PV array is between 0.096 \$/kWh and 0.109 \$/kWh. In the southern regions, the LCE of the wind turbine is between 0.083 \$/kWh and 0.093 \$/kWh, while the LCE of the PV array ranges between 0.086 \$/kWh and 0.097 \$/kWh.

In order to assess the electric grid's impact on the economic performance of the system, grid-connected HPS is compared with off-grid HPS. The case of Tozeur is considered in the comparison taking into account the optimization result found previously and all input data. The schematic diagram of the stand-alone hybrid PV/wind/battery storage system is the same as Fig. 1, with two minor changes. Instead of the grid and the

Table 9 Optimization results in each site of the country.

	PV (kW)	W	INV (kW)	Grid	TNPC ($)	COE ($)	CO_2 (kg/year)
Kebili	1.1	1	2.5	1	10,911	0.0862	1632
Thala	0.2	1	2.5	1	9224	0.0708	1776
Beja	0	1	2.5	1	7540	0.0499	1427
Elkef	0	1	2	1	8513	0.0658	1710
Jendouba	0	1	2	1	8513	0.0658	1710
Kairouan	0	1	2.5	1	7694	0.0517	1482
Kelibia	0	1	2.5	1	7687	0.0517	1471
Medenine	1	1	2.5	1	10,712	0.0843	1641
Sidi Bouzid	0.7	1	2.5	1	10,148	0.0791	1689
Siliana	0	1	2.5	1	7540	0.0499	1427
Bizerte	0	1	2.5	1	7756	0.0524	1499
Jerba	1	1	2.5	1	10,708	0.084	1639
Elborma	0.8	1	2.5	1	10,105	0.0794	1642
Gafsa	0.9	1	2.5	1	10,548	0.0829	1665
Kasserine	0.2	1	2.5	1	9225	0.0708	1787
Monastir	0	1	2.5	1	7694	0.0517	1482
Nabeul	0	1	2.5	1	7294	0.0471	1359
Tozeur	1.1	1	2.5	1	10,910	0.0862	1631
Gabes	1.4	1	2.5	1	11,466	0.0902	1625
Mahdia	0	1	2.5	1	7812	0.0532	1515
Remada	1	1	2.5	1	10,710	0.0845	1634
Sfax	0.1	1	2.5	1	9038	0.0695	1789
Tabarka	0	1	2.5	1	8513	0.0658	1710
Tataouine	1	1	2.5	1	10,704	0.0844	1632
Tunis	0	1	2.5	1	7294	0.0471	1359
Zaghouan	0.1	1	2.5	1	7411	0.0475	1331

W = wind turbine; INV = inverter; COE = cost of energy; CO_2 = carbon dioxide.

bi-directional meter, a battery bank is connected to the DC bus. A bi-directional converter is used instead of the inverter. Batteries are included as a backup system. A generic 1 kWh lithium-ion battery with a voltage of 6 V is used in the case study. Battery lifetime is considered 15 years. Each battery's capital and replacement cost is estimated at 200$, while its operation and maintenance cost is assumed at 4.4 $/year [36]. The optimal configuration of the hybrid PV/wind/battery storage system is determined using HOMER software by considering the TNPC as the main objective function.

The optimization result of the stand-alone hybrid renewable energy system is presented in Table 10. From an economic perspective, using the electrical grid as backup reduces the COE and the TNPC, respectively, by about 72% and 60% compared to the off-grid hybrid system. This result implies the importance of the electric grid on the economic performance of the renewable energy system.

From an environmental perspective, it is clear that the stand-alone hybrid renewable energy system is the best with zero CO_2 emissions per year against 1631 kg/year in the grid-connected renewable energy system. However, the consumer is primarily interested in the cost of the electricity produced.

3.2. Case study

The categorized optimal solutions given by the software for the case of Tunis are presented in Table 11. The first-row configuration with the lowest TNPC and CO_2 emissions of 1359 Kg/year is the best optimum solution. It is composed of one wind turbine, a 2.5 KW power converter, and the electrical grid. CO_2 emissions are due to the energy purchased from the grid. If we choose a combination with lower CO_2 emissions, we should add PV panels as in the second and third rows, but the TNPC of the system would increase. The third configuration becomes the best if the selling price of electricity to the grid is equal to or higher than the purchase price.

The electric power consumed by the load is 6913 KWh/year. The wind energy represents 81% of the total energy served (10,117 kWh/year), while the grid purchases represent 19% (2376 kWh/year), which corresponds to 1359 Kg/year CO_2 emissions. The total energy sold to the grid is 5074 kWh/year, with 506 kWh/year losses in the system.

The configurations that include the wind power system are more environmentally friendly, where CO_2 emissions from net-

Table 10 The optimum solution of the stand-alone hybrid renewable energy system.

Architecture				Cost		Emissions
PV (kW)	W	Battery	Converter (kW)	TNPC ($)	COE ($)	CO_2 (kg/year)
4.8	1	31	2.5	27,708	0.31	0

Table 11 Some of the optimization results.

Architecture				Cost		Emissions	PV	W	Grid	
PV (kW)	W	Grid	INV (kW)	TNPC ($)	COE ($)	CO2 (kg/year)	Production (kWh/year)	Production (kWh/year)	Energy Purchased (kWh)	Energy Sold (kWh)
0	1	1	2.5	7294	0.0471	1359	0	10,117	2376	5074
0.1	1	1	2.5	7411	0.0475	1331	147	10,117	2326	5154
0.2	1	1	2.5	7534	0.048	1303	294	10,117	2278	5223
5	0	1	4	12,939	0.0911	2305	7349	0	4031	4073

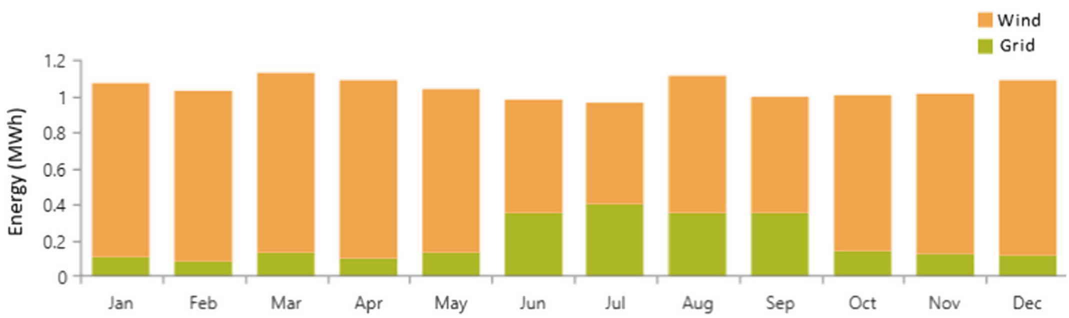

Fig. 10 Monthly average electricity production in the system.

work purchases are the lowest. This is due to the availability of wind during the day and at night, unlike solar energy.

Fig. 10 shows the electricity produced by the optimized system. The bulk of electricity purchased from the electrical grid is during the summer months due to the low wind turbine output. Indeed, the north of Tunisia is influenced by a Mediterranean climate characterized by relatively high temperatures and modest wind speed in the summer. This can be illustrated in Fig. 11, which shows the annual wind turbine power output.

3.3. Payback period

A payback period is the number of years required to recover the investment. To calculate the payback period of a HPS, a conventional base case system as a reference is required. In this study, a non-renewable power system is used as the base case system [37]. Each optimal configuration is compared with the electrical grid only to find the payback period of each case. Fig. 12 depicts the results of payback calculations for the optimal configuration of each site, which ranges between 9 and 14 years.

The project's economic benefit ranges between 11 and 16 years for each site, which proves its economic attractiveness. Fig. 13 shows the cumulative cash flow of the optimal configuration and the base case system for the site of Nabeul. The simple payback occurs when the cumulative cash flow of the difference between the base case system and optimal configuration switches from positive to negative.

The project's economic benefits will improve with the decrease in renewable energy components' price, as shown in the sensitivity analysis in the next section.

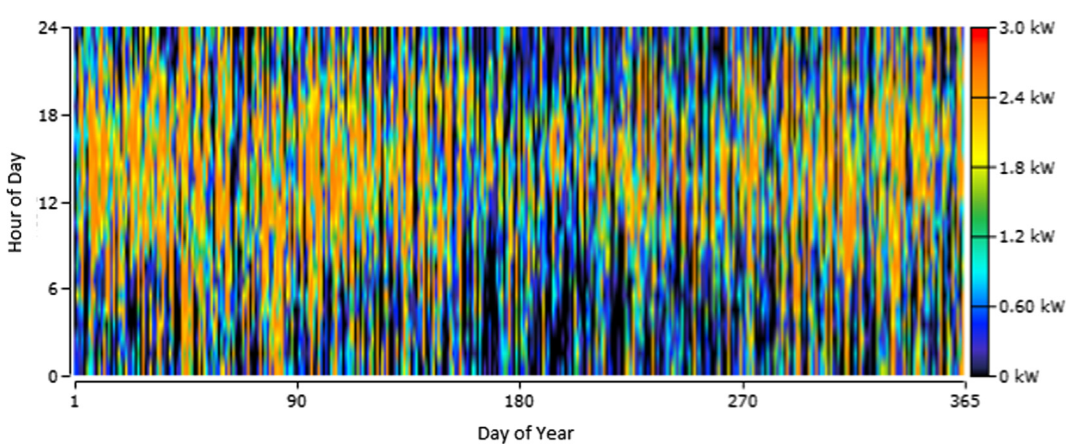

Fig. 11 Annual wind turbine power output.

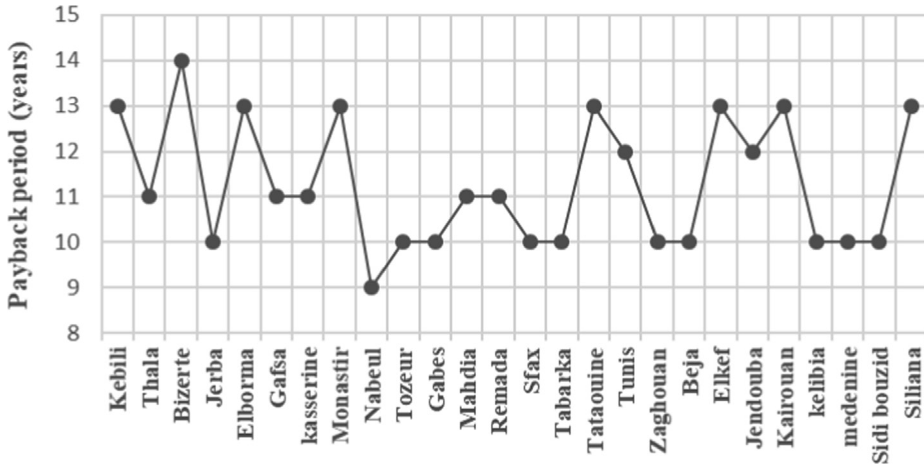

Fig. 12 Payback periods for each optimal configuration.

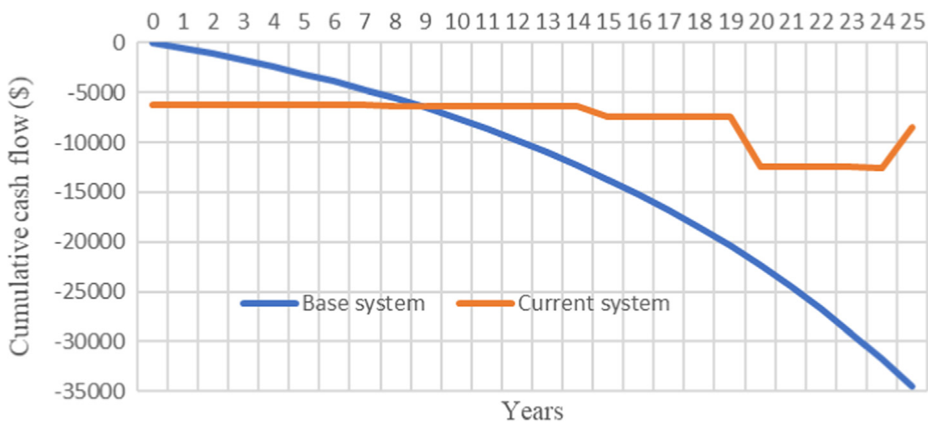

Fig. 13 The cumulative cash flow of the current and base system for the site of Nabeul.

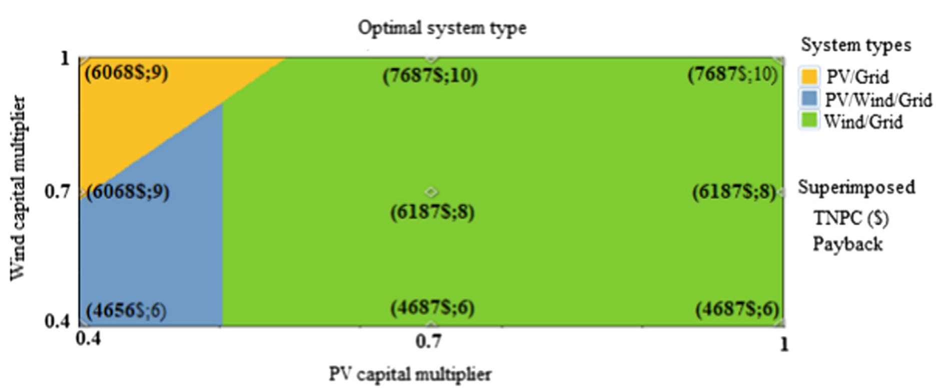

Fig. 14 Optimal system type as a function of PV and wind turbine capital multiplier variations.

3.4. Sensitivity analysis

A sensitivity analysis is used to illustrate the impacts of variation in some parameters on system performance. The International Renewable Energy Agency (IRENA) reported that the costs of renewable energy components are down and will continue to decline. IRENA noted that solar PV panels declined over 2010–2019 at 82% and onshore wind at 40% [38]. Therefore, the costs of wind turbines and PV panels are taken as sensitivity variables. In this study, the sensitivity analysis is performed for the site of Kelibia. The graphical sensitivity results are given in Fig. 14. It is show that the PV system becomes competitive if its capital cost is reduced below 57% of its current capital cost.

The lower price of renewable energy components improves the economic benefit of the project. Indeed, the TNPC and the payback period decrease as the PV panel and wind turbine cost decrease.

4. Conclusion

This paper presents a size and cost optimization of a grid-connected hybrid renewable energy system for supplying a residential load in 26 sites in Tunisia by using HOMER Pro software.

The following are the main conclusions of this paper:

- The choice of a wind turbine for a given site is an essential operation to ensure both technical feasibility and financial competitiveness in terms of high production and low electricity costs. Among the three small wind turbines selected in this work, the WES Tulipo wind turbine is the optimum choice. It has the highest capacity factor in all sites.
- Wind energy is suitable for most cities but more profitable and competitive than PV in northern and central Tunisia.
- PV system has excellent potential for development in the southern regions of Tunisia. Sfax is an exception, where wind energy is more economical than PV energy.
- The economic benefit of the renewable energy project is between 11 and 16 years during the 25-years project lifetime, which proves its economic attractiveness.
- The use of wind power in the optimal combination of the grid-connected renewable energy system is more environmentally friendly. It contributes to the reduction of CO_2 emissions, especially at night.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

[1] Renewable energy highlights, https://www.irena.org/-/media/Files/IRENA/Agency/Publication/2020/Jul/Renewable_energy_highlights_July_2020.pdf?la = en&hash = 75B114DB7A55F4260F41F64C4DFF793DB2044306.
[2] Renewable capacity statistics 2020, https://www.irena.org/publications/2020/Mar/Renewable-Capacity-Statistics-2020.
[3] Renewables 2020 global status report, https://www.ren21.net/wp-content/uploads/2019/05/gsr_2020_full_report_en.pdf.
[4] Chapter five: renewable_energy, https://rise.esmap.org/data/files/reports/rise_2018_-_renewable_energy.pdf.
[5] Projets d'énergie renouvelable en tunisie guide détaillé, http://www.tunisieindustrie.gov.tn/upload/ENR/Guide_detaille_ENR_tunisie_mai2019.pdf.
[6] http://www.steg-er.com.tn/wp-content/uploads/bsk-pdf-manager/Ministerer_de_l%27energie_14.pdf.
[7] M. Ramli, A. Hiendro, K.S. Sedraoui, Twaha Optimal sizing of grid-connected photovoltaic energy system in Saudi Arabia, Renewable Energy 75 (2015) 489–495.
[8] M. Vakilabadi, A. Afzalabadi, A. Poorfar, A. Rahbari, M. Bidi, M.T. Ahmadi, Ming Technical and economical evaluation of grid-connected renewable power generation system for a residential urban area, Int. J. Low-Carbon Tec. 14 (2019) 10–22.
[9] H.M. Sultan, A.S. Menesy, S. Kamel, A. Korashy, S.A. Almohaimeed, M. Abdel-Akher, An improved artificial ecosystem optimization algorithm for optimal configuration of a hybrid PV/WT/FC energy system, Alex. Eng. J. 60 (2021) 1001–1025. Article in press.
[10] N. Shirzadi, F.U. Nasiri, Eicker optimal configuration and sizing of an integrated renewable energy system for isolated and grid-connected microgrids, Case Urban Univ. Campus Energies 13 (2020) 3527.
[11] W. Ma, J. Fan, S. Fang, G. Liu, Techno-economic potential evaluation of small-scale grid-connected renewable power systems in China, Energy Convers. Manag. 196 (2019) 430–442.
[12] Y.Z. Hang, T. Ma, P. Campana, Y. Yamaguchi, Y. Dai, A techno-economic sizing method for grid-connected household photovoltaic battery systems, Appl. Energy 269 (2020) 115106.
[13] N. Uddin, M. Islam, Optimization and Management Strategy of Grid Connected Hybrid Renewable Energy Sources, in: International Conference on Energy and Power Engineering, 2019, pp. 190–195.
[14] S. Barakat, H.A. Ibrahim, A. Elbaset, Multi-objective optimization of grid-connected PV-wind hybrid system considering reliability, cost, and environmental aspects, Sustain. Cities Soc. 60 (2020) 102178.
[15] A.A.Z. Diab, S.I. El-ajmi, H.M. Sultan, Y.B. Hassan, Modified farmland fertility optimization algorithm for optimal design of a grid-connected hybrid renewable energy system with fuel cell storage: case study of Ataka, Egypt, Int. J. Adv. Comput. Sci. Appl. 10 (2019) 119–132.
[16] K. Ndwali, J. Njiri, E. Wanjiru, Multi-objective optimal sizing of grid connected photovoltaic batteryless system minimizing the total life cycle cost and the grid energy, Renewable Energy 128 (2020) 1256–1265.
[17] S. Ahmed, Electricity sector in Tunisia: current status and challenges—An example for a developing country, Renew. Sustain. Energy Rev. 15 (2011) 737–744.
[18] B. Brand, R. Missaoui, Multi-criteria analysis of electricity generation mix scenarios in Tunisia, Renew. Sustain Energy Rev. 39 (2014) 251–261.
[19] B. Fathi, E. Mustapha, Wind energy conversion systems adapted to the Tunisian sites, Smart Grid Renewable Energy 4 (2013) 57–68.
[20] A. Souissi, An accurate optimal sizing method of a hybrid PV/wind energy conversion system with battery storage, Int. J. Eng. Res. Afr. 48 (2020) 179–192.
[21] http://krishna.grill.free.fr/TITAN50W.pdf.
[22] Catalogue of European Urban Wind Turbine Manufacturers, http://www.urbanwind.net/pdf/CATALOGUE_V2.pdf.
[23] T. Abdali, S. Pahlavan, M. Jahangiri, A. Shamsabadi, F. Sayadi, Techno-Econo-Environmental study on the use of domestic-scale wind turbines in Iran, Energy Equip. Sys. 7 (2019) 317–338.
[24] C. Diyoke, A new approximate capacity factor method for matching wind turbines to a site: case study of Humber region, UK, Int. J. Energy Environ. Eng. 10 (2019) 451–462.
[25] Fortis Wind Energy Systems Market List Price 2018, https://www.fortiswindenergy.com/wp-content/uploads/2018/03/Fortis_Wind_Energy_Market_Pricelist_2018_SV.pdf.
[26] http://www.chinahummer.cn/index.php/index/content/175.
[27] http://www.allwindturbine.com/.

[28] http://www.steg-er.com.tn/wp-content/uploads/bsk-pdf-manager/2017-09-14_19.pdf.
[29] https://www.steg.com.tn/fr/clients_res/tarif_electricite.html.
[30] Z. Pecenak, M. Stadler, K. Fahy, Efficient multi-year economic energy planning in microgrids, Appl. Energy 255 (2019) 113771.
[31] A. Phinikarides, N. Kindyni, G. Makrides, G. Georghiou, Review of photovoltaic degradation rate methodologies, Renew. Sustain. Energy Rev. 40 (2014) 143–152.
[32] D. Jordan, S. Kurtz, K. VanSant, J. Newmiller, Compendium of photovoltaic degradation rates, Prog. Photovolt. Res. Appl. 24 (2016) 978–989.
[33] res4med country profile, https://www.res4med.org/wp-content/uploads/2017/11/Country- Profile- Tunisia-Report_05.12.2016.pdf.
[34] Tunisia electricity price, https://knoema.com/search?query = tunisia%20electricity%20price.
[35] Tunisia electricity prices, https://www.globalpetrolprices.com/Tunisia/electricity_prices/#:~:text = Tunisia%2C%20December%202019%3A%20The%20price,of%20power%2C%20distribution%20and%20taxes.
[36] M. Nurunnabi, N.K. Roy, E. Hossain, H.R. Pota, Size Optimization and sensitivity analysis of hybrid wind/PV micro-grids- a case study for Bangladesh, IEEE Access 7 (2019) 150120–150140.
[37] Homer help manual, https://www.homerenergy.com/pdf/HOMER2_2.8_HelpManual.pdf.
[38] Renewable power generation costs in 2019, https://www.irena.org/publications/2020/Jun/Renewable-Power-Costs-in-2019.